



Geospatial Data Visualization using Amazon SageMaker

Project Objectives

Cloud Infrastructure

Build a scalable, cloud-based geospatial environment using Amazon SageMaker's managed services.

Region Analysis

Implement tools to identify and analyze specific regions of interest using bounding boxes and spatial queries.

Interactive Mapping

Develop visualizations that allow users to explore geographic data through interactive web maps.

Basic Analytics

Perform fundamental spatial operations including distance calculations, area measurements, and overlay analysis.

Technology Stack

Cloud Platform

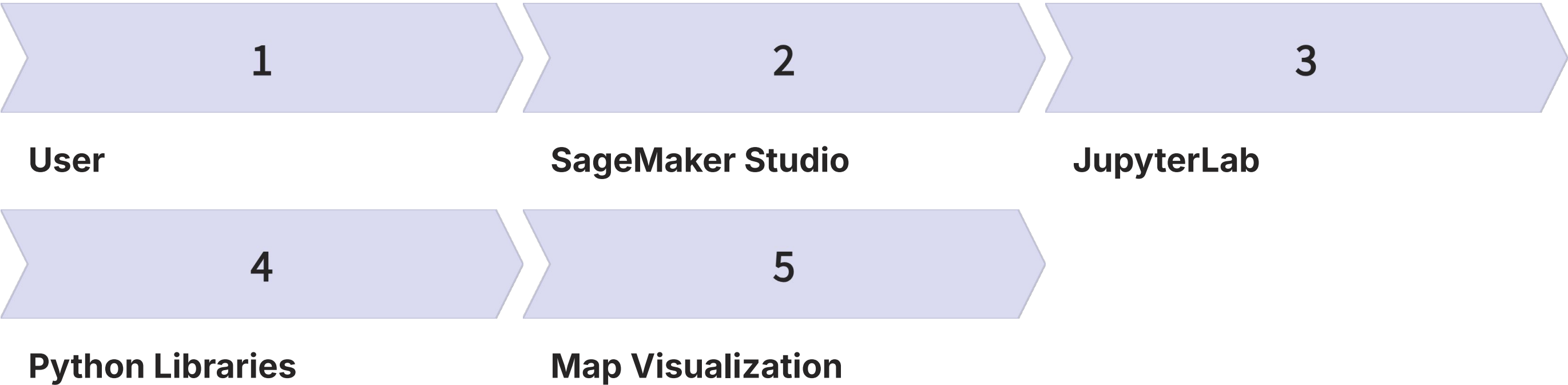
- Amazon SageMaker (managed ML platform)
- SageMaker Studio (integrated IDE)
- JupyterLab environment

Core Libraries

- Python 3.9+ runtime
- GeoPandas (spatial data manipulation)
- Folium (interactive map generation)

System Architecture

The user configures SageMaker Studio and launches JupyterLab. Via JupyterLab, Python libraries are used for map visualization.



Implementation Process



Environment Setup

Created SageMaker domain and user profile with appropriate IAM permissions for cloud resource access.



Launch JupyterLab

Accessed SageMaker Studio interface and launched JupyterLab notebook environment for interactive coding.



Library Installation

Installed GeoPandas, Folium, and related geospatial dependencies using pip package manager.

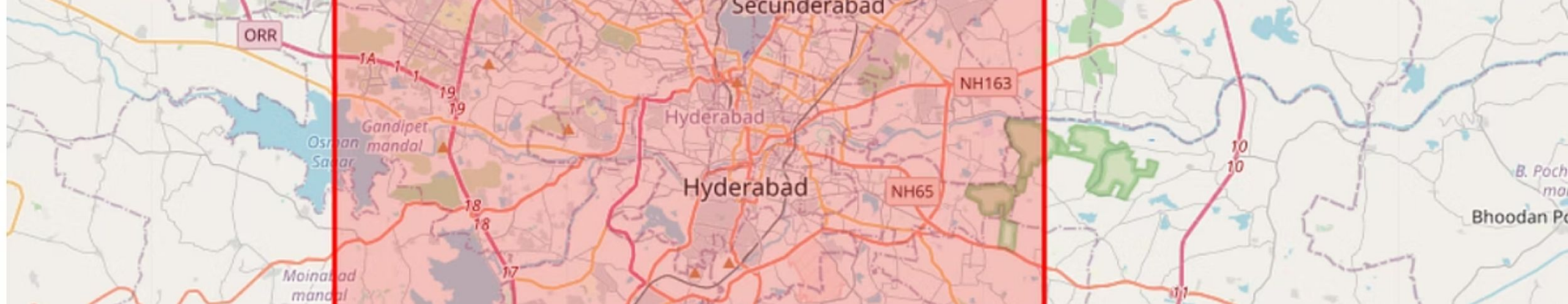


Map Development

Built interactive map visualization step-by-step, starting with base layers and adding interactive features.

Region of Interest (Bounding Box Visualization)

- Defined a geographic region using coordinate boundaries
- Visualized the selected area using a bounding box
- Highlighted a specific region for focused analysis
- Enables region-based filtering and data processing



Region of Interest (Bounding Box Visualization)

Represents how real-world geospatial systems isolate areas for analysis

Bounding Box Creation using Folium (Hyderabad Coordinates)

Defining and visualizing a geographic region using latitude and longitude

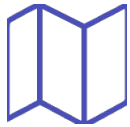
```
import folium

# Create base map (India)
m = folium.Map(location=[20.5937, 78.9629], zoom_start=5)

# Hyderabad bounding box coordinates
bounds = [
    [17.2, 78.3], # bottom-left
    [17.5, 78.6]  # top-right
]

# Add rectangle (bounding box)
folium.Rectangle(
    bounds=bounds,
    color="red",
    fill=True,
    fill_opacity=0.2
```

Key Features Implemented



Interactive Visualization

Generated web-based maps that support zooming, panning, and dynamic layer toggling through Folium integration.



Location Markers

Placed custom markers at specific geographic coordinates with popup information for data inspection.



Region Selection

Implemented bounding box functionality to define areas of interest for focused spatial analysis.



Layer Management

Created multiple map layers that can be individually enabled or disabled to compare different datasets.

Data Analysis Capabilities

Spatial Data Processing

Used GeoPandas to read, transform, and manipulate geospatial file formats including GeoJSON and Shapefile.

Region-Based Analysis

Simulated queries to identify features within specific geographic boundaries using spatial indexing.

Visualization Output

Generated choropleth maps, heatmaps, and scatter plots to represent quantitative spatial data.

Challenges & Solutions

CloudFormation Errors

Problem: Automatic stack deployment failed due to permission conflicts

Solution: Manual resource creation with granular IAM policies

Session Management

Problem: Authentication tokens expired during long-running analysis

Solution: Implemented automatic re-authentication logic

Dataset Issues

Problem: Sample dataset used in tutorial was deprecated

Solution: Sourced updated public geospatial datasets from AWS Open Data

UI Confusion

Problem: SageMaker Studio interface changed between versions

Solution: Documented new navigation paths and explored alternative workflows

Conclusion & Future Directions

Achievements

- Successfully deployed cloud-based geospatial visualization system
- Validated region-based analysis methodology
- Created reproducible notebook for team collaboration

Future Enhancements

- Integrate real-time streaming geospatial data
- Implement advanced routing and network analysis APIs
- Apply machine learning for spatial pattern detection
- Add 3D terrain visualization capabilities

